Intrusion Detection Approach for Industrial Internet of Things Traffic Using Deep Recurrent Reinforcement Learning Assisted Federated Learning

Amandeep Kaur, *Member, IEEE*

***Abstract*—The rapid growth of industrial Internet of Things (IIoT) applications generates massive amount of heterogeneous data that are prone to cyberattacks. The imperative is to secure industrial data from adversarial attacks and develop a robust and secure framework capable of withstanding sophisticated attacks. Toward this machine learning (ML) algorithms are used for intrusion detection by analyzing the devices' network traffic. However, classical ML models work on entire datasets that are located on a central server and are not a suitable choice for a secure intrusion detection framework. We propose the federated learning (FL)-based network intrusion detection model for IIoT scenarios which only share learned parameters with the central server and keep the data intact to local servers only. The proposed model is assisted with gated recurrent units (GRUs) for FL training to extract temporal dependencies of network traffic attacks in order to improve intrusion detection accuracy. Additionally, to increase the model aggregation rate of FL, we integrate deep reinforcement learning (DRL) to select of IIoT devices with high quality while keeping data privacy and energy-efficiency as main concerns. In contrast to earlier approaches, we consider non-independent and identically distributed (non-IID) data over recent IIoT datasets. Experimental findings indicate that the proposed framework outperforms state-of-the-art FL and non-FL intrusion detection models in terms of accuracy, precision, recall, F1-score, and receiver operating characterstics (ROC).**

***Impact Statement*—The deployment of intrusion detection approach for IIoT using GRU assisted federated deep recurrent reinforcement learning enhances cybersecurity posture of industrial systems in real-time scenarios. The combination of deep recurrent models and RL helps in reducing false positives, minimizing the chances of unnecessary disruptions to industrial processes that improves system's resilience and reducing the overall impact on industrial network's performance. The primary objective of this research work is to address the key challenges of industrial network and contributing to resilience, timely mitigation of potential security breaches, and uninterrupted operation of critical industrial processes, fostering a resilient and secure operational ecosystem.**

***Index Terms*—Anomaly detection, deep reinforcement learning (DRL), federated learning (FL), gated recurrent units (GRUs), industrial Internet of Things (IIoT), intrusion detection mechanism.**

Manuscript received 8 February 2024; revised 25 June 2024; accepted 9 August 2024. Date of publication 14 August 2024; date of current version 9 January 2025. This article was recommended for publication by Associate Editor Hung Nguyen upon evaluation of the reviewers' comments.

The author is with the Department of Management Studies, ABV-Indian Institute of Information Technology and Management, Gwalior, Madhya Pradesh 474015, India (e-mail: adeep5524@gmail.com).

This article has supplementary downloadable material available at https://doi.org/10.1109/TAI.2024.3443787, provided by the author.

Digital Object Identifier 10.1109/TAI.2024.3443787

I. Introduction

With the rapid growth of smart devices and next generation network technologies, the Internet of Things (IoT) gains popularity that aims to connect millions of widespread heterogeneous devices [1], [2]. The amalgamation of industrial framework with IoT is coined as Industrial IoT (IIoT) [3], [4].

The IIoT is a key technology to realize intelligent industrial systems by connecting and accessing industrial devices through internet. It has been predicted to have 36.8 billion connected IIoT connections by 2025 [5]. The IIoT devices may generate massive amount of heterogeneous data every second, susceptible to cyberattacks due to underlying connectivity with internet and exchange of sensitive data which results in disclosure of confidential information [6], [7]. These attacks can have significant consequences including data theft, data manipulation, interruption of critical operations, unauthorized control, and denial of services. Network intrusion detection system was conceptualized to cope up with specific type of attack as manifested through anomalous behavior of the network traffic. Indeed, for accurate detection of these attacks, network traffic from heterogenous IoT devices (deployed in distributed manner) has been continuously monitored which might need to share their private data [8].

To tackle the privacy challenges due to centralized learning approaches, federated learning (FL) [9] is coined as privacy-preserving decentralized machine learning (ML) paradigm that seeks the attention of academia as well as industry to develop a secure and robust framework for IIoT networks. Specifically, FL trains a model by considering the devices' local data and then aggregates the local trained models' parameters on server side located at cloud. After aggregation, the parameters of global model serve as the initial parameters for next round of local model training. As the device data is intact with local models only, FL guarantees client privacy and data security [10], [11]. A recent research work in [10] proposed FL-enabled intrusion detection in IIoT scenarios to tackle with various network and sensor attack categories by deploying differential privacy technique and validated on TON_IoT and CIC-IDS2018 datasets.

However, some recent studies pointed that traditional FL faces heterogeneity issues as it assumes that all the heterogenous data sources contribute equally to train model parameters. But

2691-4581 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

this assumption degrades the performance of collaborative training as the generated data is highly skewed and nonindependent and identically distributed (non-IID). To tackle this issue, deep reinforcement learning (DRL)-based FL framework was adopted to select IIoT devices based on certain criterion such as data quantity, data quality and data type, to enhance the model aggregation rate [12], [13]. Zhang et al. [13] proposed a DRL assisted FL in IIoT environment for the selection of high-quality participating nodes while maintaining the privacy and heterogeneity of the data generated by the nodes. Thus, DRL enable nodes to dynamically adapt their participation in FL based on changing conditions, ensuring that the nodes with high quality contribute in model aggregation.

While the use of DRL with FL has been considered by various researchers [14], [15] for the improvement of data aggregation process but state-of-the-art work did not analyze the impact for intrusion detection, and do not address the impact of non-IID data distribution to estimate real-world scenarios. To address this gap, this study proposed FL based on DRL for network intrusion detection framework in IIoT network. Unlike previous studies, we assess the performance of proposed framework across three datasets, consist of recent attacks related to IIoT scenarios and non-IID data distribution. Additionally, the intrusion detection environment is dynamic with threats evolve over the period of time. Thus, gated recurrent units (GRUs) are combined with FL process which adapt to new attack strategies and changing network attack patterns by mapping their temporal dependencies which enhances the performance of proposed model further. In summary, the key contributions of this research work are as follows:

1) We introduce a novel GRU-assisted FL-based network intrusion detection framework for IIoT networks. The proposed framework maintains data privacy through decentralized local training, improves detection accuracy via collaborative model aggregation, and provide scalable, and resource efficient security monitoring in industrial environment.
2) We integrate DRL into the framework to select high quality IIoT nodes, ensuring efficient and effective participation of nodes, which further enhances the performance and robustness of intrusion detection system.
3) We conduct a comprehensive evaluation of proposed framework on diverse IoT datasets including ToN-IoT, Edge_IIoT, and X-IIoTID, considering non-IID data distributions. The experimental results demonstrate its robustness in handling real-world data complexities and significant reductions in energy consumption compared to traditional FL approaches, highlighting its practical applicability in real-world IIoT scenarios.

The rest of the article is organized as follows. Section II briefly outline the literature review. In Section III, we describe the FL-DRL framework for device selection. The proposed methodology is detailed in Section IV. Section V presents the performance analysis of proposed framework. Finally, Section VI presents concluding remarks and outlines future directions.

II. Related Works

In recent years, the research community has perceived growing interest in developing a secure and robust network intrusion detection framework to protect IIoT data against malicious cyberattacks. Certain approaches based on ML and deep learning (DL) has been investigated in literature for intrusion detection. For instance, Vu et al. [16] proposed supervised learning-based anomaly detection model in IoT. Specifically, the authors developed regularized versions of autoencoders to extract latent features of input data to detect unknown attacks. Further, Long et al. [17] developed semisupervised learning-based ladder network model for intrusion detection in IIoT network. The performance of this network was evaluated on CIC-IDS2018 dataset. Additionally, Hansan et al. [18] utilized various ML algorithms for anomaly detection in IoT network. Lin et al. [19] proposed secure data fusion strategy with transfer learning for IIoT scenarios. Ren et al. [20] developed a DRL-based feature selection framework for intrusion detection mechanism and validated on CIC-IDS2018 dataset only. Zhang et al. [21] proposed graph neural network-based learning model for intrusion detection framework. The model overcome the difficulties such as data nonlinearity and high-dimensional data but limits its performance due to high computational requirements. In addition, Li et al. [22] presented DL-based intrusion detection mechanism as multiclassification technique using multi-convolutional neural network (CNN) fusion for IIoT. The proposed mechanism divides the features based on correlation, converted into grayscale graph and assessed the performance using NSL-KDD dataset. Also, Alsaedi et al. [23] implemented supervised ML models in IoT/IIoT applications for intrusion detection and assessed the performance using TON_IoT dataset. Several research works also concentrate to extract temporal and spatial characteristics of network traffic so as to improve detection accuracy and reduce the false alarm rate. In this context, Halbouni et al. [24] developed a CNN–long-short term memory (LSTM)-based hybrid intrusion detection model. The proposed framework was trained and evaluated using CIC-IDS2017, UNSW-NB15, and WSN-DS datasets. Li et al. [25] proposed bi-LSTM network along with multifeature layer to extract corresponding attack features from historical data. The proposed model demonstrated the improved efficiency to detect poisoning attacks in network. Similarly, Kong et al. [26] proposed an integrated framework using generative adversarial network (GAN) with Bi-LSTM to capture time series dependence of attack history. Additionally, Latif et al. [27] proposed recurrent neural networks (RNNs)-based intrusion detection mechanism. Several authors reported GRU-based prediction models as intrusion detection framework in industrial cyber-physical systems [28]. However, these works did not consider the data security aspect while considering their intrusion detection approaches for IIoT scenarios.

In this context, several researchers developed an intrusion detection framework leveraging FL. For example, Wang et al. [29] proposed FL-based anomaly detection in IIoT scenarios with each local model trained by DRL to reduce privacy leakage. In addition, Wang et al. [43] also proposed Hierarchical FL for trajectory anomaly detection in intelligent vehicles. Li et al. [30] proposed DeepFed framework, with advanced encryption

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

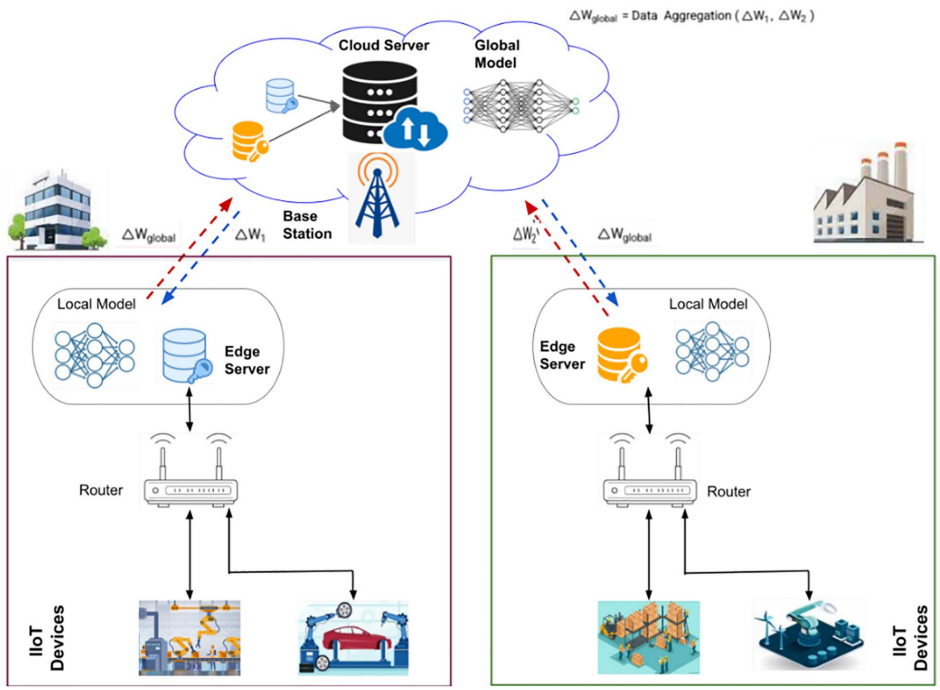

Fig. 1. Proposed FL framework for IIoT.

standard, to ensure the security during the FL training process. The integrated framework with CNN and GRU were used to detect various threats in industrial networks. Further, the research study in [31] proposed FL-based attention GRU (FedAGRU) in wireless networks. The performance was evaluated using datasets namely KDD CUP 99, CIC-IDS2017, and WSN-DS. Weinger et al. [32] proposed FL-based model to imbalanced dataset using random oversampling and SMOTE process. Additionally, GANs was utilized for data augmentation for improving detection performance. Furthermore, Liu et al. [33] proposed an attention mechanism integrated with CNN and LSTM models to effectively learn fine-grained features of time series data, aimed at enhancing anomaly detection within an IIoT network.

In a nutshell, these research works consider the data privacy aspect with FL for intrusion detection in IIoT scenarios and evaluated their performance using various datasets. However, these frameworks considered various IIoT/IoT nodes equally for the aggregation process of FL training which limits the model's performance. Moreover, the performance was evaluated using non-IIoT specific datasets only. Moreover, to improve the performance of FL aggregation process specifically as intrusion detection mechanism, limited literature is available. For instance, authors in [34] developed an attention mechanism to determine the attention value of each participating device corresponding to variation in distribution of data. This work did not consider the communication cost incurred during FL training process and temporal dependencies of network traffic. Also, the performance was evaluated using NSL-KDD and cloud-based ISOT-CID datasets only.

In contrast to works cited above, we propose GRU-assisted FL intrusion detection framework for IIoT scenarios. The proposed framework is evaluated using datasets which specifically includes recent attacks categories from IIoT network, including protocol-related attacks, and data manipulation attacks. As of now, to the best of our knowledge, there is no secure intrusion detection model specifically designed for IIoT networks that achieves improved accuracy in intrusion detection, enhances aggregation process, and reduced communication costs compared to existing FL-based solutions. Further, the performance is evaluated using IIoT attack-specific datasets by considering non-IID data distribution highlighting its practical applicability in real-world scenarios.

III. Federated DRL Framework

A. System Model

We consider industrial scenario with heterogeneous IIoT devices, cloud layer and the edge layer. In this scenario, the attackers tie themselves to IIoT ecosystem via local servers/edge servers in order to establish communication with deployed sensors. Similar to edge server, attacker server also receives data, behaves similar to the edge server to divert the data from edge server to its own server. Fig. 1 depicts the system model of FL-based cloud IIoT framework which consist of multiple deployed sensors in industrial environment, edge servers and cloud server to handle heterogeneous high-volume data. As illustrated in Fig. 1, the data is collected from deployed devices (i.e., sensors and actuators) in industrial firms. These devices collect information at specific interval of time and sent to the respective edge server of industrial firm for training process of local models. Here, the proposed network intrusion detection model deployed at router to detect malicious activities by analyzing network traffic patterns or anomalies that may specify an intrusion or suspicious behavior. All these devices are connected using Wi-Fi. The edge servers and cloud server are communicated using high-speed internet. At cloud, the global model aggregates the parameter updates received from local models, update its parameters, and sends back to associated edge servers to retain local models. The

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

whole process continuous till the local model converges and best to detect the network traffic intrusion [10].

Basically, FL process involves computation and communication process. The computation process includes local model updates of participating devices whereas communication process involves the exchange of model parameter or updates between devices and cloud server. The energy consumption during communication and computation process in FL process is as follows.

1) *Energy Consumption Model:* The energy consumed during FL training process includes the energy consumption to collect data from local devices, the computation of local training and the communication process for global aggregation [11], [35]. Let $i \in [1, 2, ..., N]$ be the IIoT devices, the central processing unit (CPU) energy consumption of each IIoT device i to collect and preprocess device data during local round of computation be expressed as

$$E_i^{\text{data}} = D_i \varsigma_i \upsilon_i^2 \tag{1}$$

where $D_i > 0$ denotes the size of local dataset generated by device i, ς_i denotes number of CPU cycles and $\upsilon_i < \upsilon_{\max}$ indicates the frequency of CPU cycle, which is constrained within feasible reasons of CPU frequency.

Further, the energy consumed for computation of local training dataset of device i can be expressed as [36]

$$E_i^{\text{comp}} = D_i \varsigma_i K_{L,i} m_i \upsilon_i^2 \tag{2}$$

where $K_{L,i}$ is number of computational rounds for local training and m_i represents model size. Finally, to upload the parameters from edge server to the cloud server, the communication energy consumption can be written as

$$E_i^{\text{comm}} = \frac{\rho q}{\sum_{u=1}^{U} t'_{i,u} B_i log_2(1 + \zeta_{i,u})} \tag{3}$$

where q denotes number of bits quantized before transmission, ρ represents normalization for communication energy efficiency, B_i is subchannel bandwidth where B_i satisfies $\sum_{i=1}^{U} B_i \leq B_{\max}$, and $t'_{i,u}$ indicates the time fraction the device i occupies subchannel u. The signal to noise ratio (SNR) is calculated as $\zeta_{i,u} = (P_{i,u} g_{i,u} / N_0)$ where $P_{i,u}$ and $g_{i,u}$ denotes transmission power and channel power gain, and N_0 refers to the noise power. Thus, the total energy cost (or average energy consumption) involved in FL process is as follows:

$$C_{\text{training}}^t = \frac{\sum_{i=1}^{N} \left(E_i^{\text{data}} + E_i^{\text{comp}} + E_i^{\text{comm}}\right)}{N}. \tag{4}$$

2) *Data Quality Model:* The selection state of the device i is evaluated based on the contribution made by the device to improve the global model parameters by training with local model parameters in τ round of training, which is calculated as

$$C_{dq}^t = \frac{1}{\tau} \sum_{\tau} \frac{w_\tau^i - w_\tau^G}{\Delta_\tau} \tag{5}$$

where w_τ^i and w_τ^G represents the local model and global model parameters of device i after τ th round of training, Δ_τ represents the sum of weights of all contributing devices after the τ th round of training.

B. Selection of FL Client Nodes Selection With DRL

The selection of nodes for IIoT equipment is a combinatorial optimization problem. To solve this problem, we transform it into Markov decision process (MDP), denoted by $M = \{\text{S}, \text{A}, R, P, \gamma\}$, where S is state space, A denotes the action space, R signifies the reward, P indicates the state transition probability and γ is discount factor. Specifically, the learning agent learns from the past interactions, obtain optimal decisions during each time iteration until the convergence to the optimal strategy $\pi^*(s)$. The parameters of the MDP model are defines as follows.

1) *State Space:* Initially, the agent collects the information in the form of state corresponding to each node by interacting in an environment. The current state $s_t \in \text{S}$ in time slot t, is expressed as follows:

$$s_t = \{\varsigma_t, \upsilon_t, \kappa_{t-1}, \zeta_t, C_{dq}^t\} \tag{6}$$

where ς_t and υ_t indicates available computing resources, κ_{t-1} represents selection state of devices in previous time slots, ζ_t indicates current value of SNR and C_{dq}^t represents data quality.

2) *Action Space:* The agent action $a_t \in \text{A}$ is the selection of device which is expressed as follows:

$$a_t = (\kappa_i^t, P_{i,u}) \tag{7}$$

where $\kappa_i^t = 1$ indicates the selection of device i for participation in training with transmission power $P_{i,u}$, and $\kappa_i^t = 0$ otherwise. The optimal transmission power is adjusted based on channel conditions (measured as SNR value).

3) *Reward:* The reward function assesses the action taken by an agent within a given state s_t defined as follows:

$$r(s_t, a_t) = -\left\{\frac{\sum_{i=1}^{i_s} \left(C_{\text{training}}^t + C_{dq}^t\right).a_i^t}{\sum_{i=1}^{i_s} a_i^t}\right\} \tag{8}$$

where $i_s \in [N]$ represents the selected devices and a_i^t indicates action as the selection of device i.

4) *Policy:* The policy denotes the mapping of state into action $\pi : S \rightarrow A$. In time slot t, the executed action can be derived from the policy with expected return value expressed as

$$\vartheta_\pi(s) = \text{E}_\pi[R(s_t, a_t) + \gamma \vartheta_\pi(s_{t+1}) \mid s_t = s], \forall s \in \text{S}, a \in \text{A} \tag{9}$$

where $\gamma \in (0, 1]$ represents reward discount factor that cares about the future long-term reward [37]. To achieve an optimal policy $\pi^*(s)$, it must satisfy Bellman's optimality equation as $\vartheta_{\pi*}(s) = \max_a \vartheta_\pi(s)$.

Various RL methods can estimate the optimal policy, but Q-learning (QL) is among the most widely used model-free approaches. It does not necessitate knowledge of the transition dynamics between states. The optimal Q-value $\vartheta_{\pi*}(s)$ for all state-action pairs can be determined as

$$Q_{\pi*}(s) = R(s_t, a_t) + \gamma \text{E}_\pi[\vartheta_\pi(s_{t+1})], \forall s \in \text{S}, a \in \text{A} \tag{10}$$

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

where $\vartheta_{\pi*}(s) = \max_a\{Q_{\pi*}(s,a)\}$. The optimal Q-value can be obtained in recursively by

$$Q_{t+1}(s_t,a_t) = Q_t(s_t,a_t) + \alpha[R(s_t,a_t) + \gamma \max_{\hat{a}} Q_{t+1}(s_{t+1},\hat{a}) - Q_t(s_t,a_t)]. \tag{11}$$

Here, $\alpha \in (0,1]$ indicates learning rate for DRL process. In learning process, each learning agent initialize a Q-table to store all state-action pairs and updates according to (11). The QL algorithm converges to an optimal policy given sufficient learning time in small state-action spaces. However, as the size of the state-action space increases, it becomes impractical. Constructing a Q-table for all possible state-action combinations requires immense memory, leading to infeasible storage requirements and significant delays in convergence [38].

Specifically, DRL algorithm utilizes neural network that estimates the state instead of mapping every solution as state-action pairs, creating a manageable solution space by storing weight θ which is typically small in size. In DRL algorithm, an action is selected based on two neural nets: primary and target neural network, along with experience replay ξ. During the learning process, a mini-batch $\xi \sim U(D)$ is uniformly drawn to update the neural networks. This process provides improved efficiency as neural networks are trained by same training experience over several iterations. The mini-batch is inputted to both primary and target networks. The Q-network considers the current state-action pair from each sample of mini-batch to obtain "Predicted Q-value(Q_{net})" corresponding to that particular action. The target network further examines the next state from each mini-batch sample to forecast the highest Q-value for all potential actions, aiming to determine the "Target Q-value $(\hat{Q}_{\text{net}})$" which are utilized to compute the loss function as

$$\ell_\phi(\theta_\phi) = \mathrm{E}_{e_l \sim U(D)}\left[\left(\begin{array}{c} R(s_l,a_l) + \gamma \max_{\hat{a}\in \boldsymbol{A}} \hat{Q}_{\text{net}}(s_{l+1},\hat{a};\theta_\phi^-) \\ -Q_{\text{net}}(s_l,a_l;\theta_\phi) \end{array}\right)^2\right] \tag{12}$$

where θ_ϕ^- and θ_ϕ are the weight parameters of $\hat{Q}_{\text{net}}$ and Q_{net}, respectively. To minimize this predicted loss function, gradient descent is employed for updating as

$$\theta_\phi^- \leftarrow \theta_\phi - \left(\begin{array}{c} R(s_l,a_l) + \gamma \max_{\hat{a}\in \boldsymbol{A}} \hat{Q}_{\text{net}}(s_{l+1},\hat{a};\theta_\phi^-) \\ -Q_{\text{net}}(s_l,a_l;\theta_\phi) \end{array}\right) \nabla_{\theta_\phi} Q_{\text{net}}(s_l,a_l;\theta_\phi). \tag{13}$$

The process repeats for the next time-steps to update the parameters of neural network until its convergence.

C. FL-DRL Framework

To enhance the convergence rate and accuracy of the FL model, a device with high quality and minimal overall cost is chosen to participate in the training process. Initially, the cloud server initiates the FL process by broadcasting the global model and initialization parameters. Subsequently, the server selects the optimal subset of participating devices through DRL process. Let us consider the selected participating nodes $i_s \in N$ having local datasets as $D = \{D_1, D_2, ...D_f, ..., D_{i_s}\}$ then the edge server trains the local model for e epochs with batch size Λ to calculate loss function as

$$L_f(w_f) = \frac{1}{D_f}\sum_{f=1}^{i_s} \ell_f(w_f) \tag{14}$$

and the local problem is

$$w_f^t = \arg\min_{w_f} L_f(w_f \mid w_f^{t-1}), w_f \notin \mathbb{R}. \tag{15}$$

Subsequently, the global loss function is calculated as

$$L(w^t) = \frac{\sum_{f=1}^{i_s} D_f L_f(w_f^t)}{\Delta} \tag{16}$$

where $\Delta = \sum_{f=1}^{i_s} D_f$. During each training round, the update rule with Stochastic gradient descent (SGD) is as follows:

$$w_f^{t+1} = w_f^t - \eta \nabla L_f(w_f^t) \tag{17}$$

where $\eta \geq 0$ represents hyperlearning rate. After τ learning rounds, the global parameters are updated at the cloud server for aggregation expressed as

$$w^{t+1} = \frac{\sum_{f=1}^{i_s} D_f w_f^{t+1}}{\Delta}. \tag{18}$$

The updated global parameters w^{t+1} are employed for the next iteration of the training process.

IV. Proposed GRU-Assisted Federated Deep Recurrent Reinforcement Learning Framework for Intrusion Detection

Within this section, we introduce a novel framework aimed at detecting anomalies in IIoT networks. This framework utilizes a GRU-assisted FL-DRL approach. Deploying an intrusion detection model in IIoT nodes is quite challenging task due to limited computing resources and heterogeneity of network traffic data. However, once the high-quality local devices are selected with DRL, the proposed framework deals with network intrusion detection system which is deployed on router to continuously monitor the network traffic behavior to identify possible cyber-attacks or malicious activities.

A. Data Preparation

The raw network traffic undergoes the process of data preparation before it is fed in the proposed framework. This process often removes unwanted features, converts nonnumerical features and handling missing values, if any to reduce memory requirements and computational cost.

1) *Data Preprocessing:* The collected data includes categorical features of each IoT and IIoT sensors having nominal values are converted into numeric fields. To represent categorical or nonnumerical parameters in a format suitable for ML models, a technique known as one-hot encoding is employed. This method transforms

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

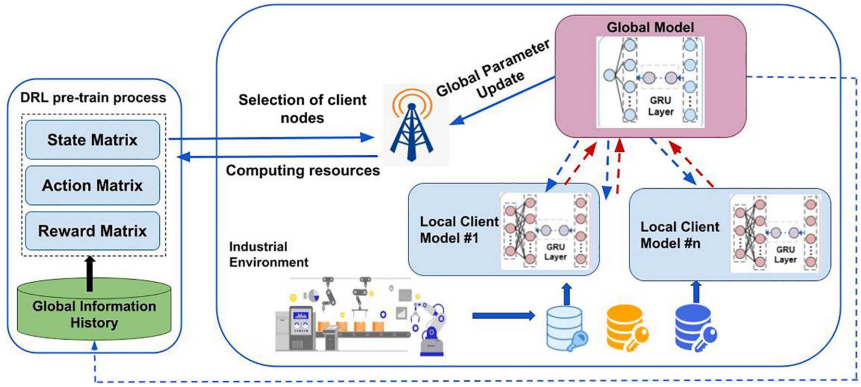

Fig. 2. Schematic representation of the proposed GRU-assisted federated deep recurrent reinforcement learning framework.

the original values into a series of binary representations, where each unique value is assigned a separate column or feature with a value of either 0 or 1.

2) *Data Normalization:* It is a crucial step that helps ensure all features contribute equally to the learning process. By normalizing the data, potential biases introduced by disproportionately large feature values are mitigated, leading to a more balanced and fair model training process. A max–min normalization method is used to scale feature values as

$$z = \frac{(y - y_{\min})}{(y_{\max} - y_{\min})} \tag{19}$$

where $y_{\max}$ and $y_{\min}$ represents the maximum and minimum values, respectively.

B. Proposed GRU-Assisted Federated Deep Recurrent Reinforcement Learning Framework

To develop a secure framework for the protection of heterogeneous industrial data from malicious attacks, we built a novel GRU-assisted FL architecture. GRU incorporates recurrent connections between neural network layers, enabling it to store pertinent state information in cell units, thereby capturing temporal relationships inherent in network traffic data. This feature renders GRU suitable for predicting sequential data. Fig. 2 illustrates the complete workflow of the proposed framework.

Under the considered setting, each local dataset D_{i_s} consist of input-output pairs $\{x_i, y_i\}_{i=1}^{|D_{i_s}|}$. The input sample vector $x_i \in \mathbb{R}^d$ with d features having labeled output value $y_i \in \mathbb{N}$. For input training sample x_i, we need to find the model parameter vector w with loss function $\ell_i(w)$ that characterizes the output y_i is expressed as follows:

$$\ell_i(w) = \frac{1}{2}(x_i^T w - y_i). \tag{20}$$

When GRUs are integrated with FL, each IIoT device locally trains the GRU model on its own with local dataset and learns the temporal dependencies effectively to distinguish normal traffic from potential intrusions. The architecture of the proposed intrusion detection model includes input layer, two GRU layers and one fully connected layer. A typical GRU structure includes reset gate, update gate, hidden state, and current input state. The update gate keeps the useful information for long time and remove the irrelevant information using reset gate. After local training, each device shares its trained GRU model parameters with a central server, adhering to privacy constrains. Further, to implement a secure mechanism for parameter aggregation that minimizes the following loss function in the framework can be written as follows:

$$J_i(w) = \frac{1}{|D_{i_s}|} \sum_{i=1}^{|D_{i_s}|} \ell_i(w) + \lambda h(w) \tag{21}$$

where $\lambda \in [0, 1]$ and $h(.)$ represents the regularizer function. At the aggregation server, the global predicted model can be represented as follows [39].

$$\arg\min_{w \in \mathbb{R}^d} J(w) \tag{22}$$

where

$$J(w) \equiv \frac{\sum_{i=1}^{|D_{i_s}|} D_i J_i(w)}{\Delta} \equiv \frac{\sum_{i=1}^{|D_{i_s}|} \ell_i(w) + \lambda h(w)}{\Delta}. \tag{23}$$

The ultimate goal of this structure is to minimize the error between the actual and predicted value as loss function. During each learning round, local model is updated as

$$w_{r+1}^{i_s} = w_r^{i_s} - \eta \nabla J_{i_s}(w_r^{i_s}). \tag{24}$$

At cloud server, the aggregation of parameters generated by local updates are combined as

$$w^{\mathrm{G}} = \sum_{i_s} \frac{D_{i_s} w^{i_s}}{\Delta}. \tag{25}$$

C. Intrusion Detection Mechanism

Considering the scenario from heterogenous industrial equipment's which are connected IIoT devices as illustrated in Fig. 1. The data generated by these equipment's is stored at edge server of associated industrial firm, that will be used to train the local model in FL training process. Each local model employs GRUs to capture temporal dependencies in the network traffic data. Algorithm 1 describes the pseudocode of proposed learning model which initially collects data, define local variables required in the procedure, such as network parameters, hidden layers,

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: GRU-Assisted Federated Deep Recurrent Rein-Forcement Learning

Input: Number of rounds τ, number of Epochs e, total IIoT devices i, hyperlearning rate η, batch size Λ, local communication rounds K_L

```
Initialize all units of GRU network,
for each round τ do
Let x and y be the input and their corresponding labels in local dataset
{i_s} ← select participating IIoT devices for FL training with DRL
Edge server broadcast w_r^G to selected IIoT devices
    for each selected device i_s ∈ N in parallel do
        for each epoch e do
            Sample mini-batch dataset {x_m, y_m}_{m=1}^{Λ} from dataset
            Compute loss function according to (20) and (21)
            Update model parameters w_{r+1}^{i_s} = w_r^{i_s} − η∇J_{i_s}(w_r^{i_s})
            while global model has not converged do
                Send w_{r+1}^{i_s} to cloud server
                Perform model aggregation according to (25)
            end
        end for
    end for
Cloud server disperses updated global model for next round of training
Update corresponding weights of GRU network
end for
```

batch size, learning rate, discount factor, and global model parameters. A loop is executed for τ rounds, select participating devices, and collect data from these nodes at respective edge server. After data preparation of collected data, the local models are trained for intrusion detection at edge server. The parameters of these local models after training are set to cloud server for updating the parameters of global model. The iterative process involves the redistribution of an updated global model from a cloud server to edge servers located at various industrial firms for next round of training. At these edge servers, the local models continue their training process using site-specific data until they achieve a high accuracy in detecting intrusions or anomalies within the incoming network traffic. This cycle of updating the global model, disseminating it to edge locations, and refining the local models through further training is repeated until the desired accuracy are met across all devices in industrial networks. The network model is fed with the collection of data samples called mini-batch and weights are updated after each epoch. To achieve significant performance of proposed model, we evaluate the performance with different learning parameters and batch size.

D. Convergence Analysis

Before considering the convergence of Algorithm 1, following assumptions are made, which make each global update behave like SGD update. This implies that convergence of proposed model should not differ too much from SGD [38] under following assumptions:

1) *A1: For each decision epoch i, $\ell_i(w)$ is strongly convex*

$$\ell_i(w') \geq \ell_i(w) + \nabla\ell_i(w).(w\prime - w)^T + \frac{\mu}{2}\|w - w'\|_2^2 \quad \textit{for all } w \textit{ and } w' \tag{26}$$

2) *A2: The loss function in FL training satisfies L-smooth,*

$$\ell_i(w') \leq \ell_i(w) + \nabla\ell_i(w).(w' - w)^T + \frac{L}{2}\|w - w'\|_2^2 \quad \textit{for all } w \textit{ and } w' \tag{27}$$

3) *A3: Let ξ_t^i be the random sample from* ith *local device's data uniformly with variance of stochastic gradients is bounded and there exists $\sigma > 0$* such that

$$\left\|\nabla\ell_i(w_t^i, \xi_t^i) - \nabla\ell_i(w_t^i)\right\|^2 \leq \sigma_i^2 \quad \text{for } i = 1, 2, \ldots, N \tag{28}$$

4) *A4: The expected squared norm of stochastic gradients is uniformly bounded for constant G* as

$$\mathrm{E}\left\|\nabla\ell_i(w_t^i, \xi_t^i)\right\|^2 \leq G^2 \text{ for } i = 1, 2, \ldots, N \text{ and } t = 0, \ldots, T-1 \tag{29}$$

5) *A5: Assume $S_t \subset [N]$ contains subset of selected participating device $i_s \in S_t$* selected with probability $p_1, \ldots, p_N$, then the aggregation process performs $w_t \leftarrow (1/i_s)\sum_{i_s \in S_t} w_t^{i_s}$.
6) Let ℓ^* and ℓ_i^* denotes their minimum values. The term $\Gamma = \ell^* - \sum_{i=1}^{N} p_i \ell_i^*$ quantifies the magnitude of non-IID to reflect the heterogeneity of data distribution. In case of IID, Γ assumed to be 0.

Theorem 1: Let the Assumptions 1 to 5 hold, choose $\iota = (L/\mu)$, $g = \max\{8\iota, \tau\}$, learning rate $\eta = (2/\mu(g+t))$ and $C = (4/i_s)\tau^2 G^2$ then

$$\mathrm{E}[\ell(w_T)] - \ell^* \leq \frac{2\iota}{g+T}\left(\frac{B+C}{\mu} + 2L\|w_0 - w^*\|^2\right) \tag{30}$$

where $B = \sum_{i_s=1}^{S_i} p_{i_s}^2 \sigma_{i_s}^2 + 6L\Gamma + 8(\tau - 1)^2 G^2$

Proof. For detailed proof, please refer Appendix A, supplementary material.

V. Performance Evaluation

A. Experimental Setting

We implemented the proposed model using Python 3.6. The key libraries utilized included Keras, Numpy, Scikit-Learn, and TensorFlow. All experiments ran on a workstation equipped with an Intel® core i9- 12900K CPU, NVIDIA GeForce RTX 4090 GPU 24 GB, and 64-bit operating system (Windows 11 Professional) listed in Table I. All other parameters including learning parameters of proposed framework are listed in Table II.

1) Datasets and Evaluation Metrics: We assess and validate the performance of the proposed intrusion detection model by utilizing three datasets: TON_IoT, Edge-IIoT, and X-IIoTID. The details are as follows:

a) *TON_IoT [23]*: The database is specifically designed to evaluate IIoT defense solutions. It consists of variety of

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CONFIGURATION OF EXPERIMENTAL SETUP

System Setup	Specification
CPU	Intel(R) core i9
GPU	NVIDIA GeForce RTX 4090 24 GB
OS	Windows 11 Professional
RAM	128 GB
Implementation platform	Python
Libraries	Keras, Numpy, Scikit-Learn, TensorFlow

TABLE II
HYPERPARAMETERS OF PROPOSED MODEL

Parameter	Value	Parameter	Value
GRU model		**FL**	
Number of layers	2	Number of rounds	100
Hidden units	128	Local epochs	5
Dropout rate	0.3	**DRL**	
Optimizer	Adam	Replay buffer size	10 000

IoT related attacks and legitimate instances. It contains 46 labeled features as *normal* and *attack* to classify various observations and type of features to indicate subclass of each attack as multiclass classification problem. A common type of cyberattacks in TON_IoT are as follows.

i) *Scanning:* It is commonly used by cyber attackers to gather information about a target network. It involves actively probing the target to uncover potential entry points, identify vulnerabilities, and locate valuable resources.

ii) *DoS:* It is well known flooding attack in which attacker floods the target with a tremendous volume of data or connection requests, exhausting its resources such as bandwidth, memory, or processing power, with an objective to disrupt the normal functioning and accessibility of the targeted resource.

iii) *DDoS:* In this attack, the target IIoT device is flooded with numerous numbers of connections in order to exhaust the resources of the target device.

iv) *Ransomware:* It is designed to deny a user access to the system or services by encrypting them and then provides access back by selling a decryption key.

v) *Backdoor:* Backdoor attack bypasses the systems' customary security mechanisms with backdoor malware.

vi) *Injection Attack:* It manipulates data as well as control commands of IIoT system and also disrupts its normal operation by injecting malicious data/codes.

vii) *XSS:* It often attempts to inject malicious scripts into web pages viewed by other users to perform various malicious activities.

viii) *Password Violations:* The attacker aimed specifically to obtain password by using variety of methods including password cracking, password sniffing, dictionary attack, or guessing password via brute search.

ix) *MITM Attack:* This attack involves an unauthorized third-party intercepting and eavesdropping on communications between two parties who believe they are communicating directly with each other.

TABLE III
TRAINING AND TEST SAMPLES OF TON_IOT DATASET

Attack Category	Training Dataset	Test Dataset
Normal	245 000	55 000
Scanning	20 000	3973
DoS	20 000	4000
DDoS	20 000	4000
Ransomware	16 030	3970
Backdoor	20 000	4000
Injection attack	20 000	4000
XSS	13 844	6116
Password violations	20 000	4000
MITM	593	459

TABLE IV
TRAINING AND TEST SAMPLES OF EDGE-IOT DATASET

Attack Category	Training Dataset	Test Dataset
Normal	19 281	4820
Scanning	15 869	3967
DDoS	38 669	9667
Malware	23 621	5905
Injection	24 030	6009
MITM	286	72

Table III provides a breakdown of the number of records belonging to each attack category within the TON_IoT dataset, which were considered for the evaluation process.

a) *Edge_IIoT [41]*: Edge-IIoT dataset collects the data from sophisticated seven-layer testbed (in federated architecture). It includes 10 IoT devices and IIoT-based ModBus communication flows. It also includes 14 different types of attacks targeting IoT and IIoT protocols, which are categorized into five distinct threat categories, namely, DDoS, Scanning, Malware, Injection and MITM attacks similar to TON_IoT dataset. For evaluation, Table IV illustrates distribution of normal and attack categories for training and test sets.

b) *X-IIoTID [42]*: The dataset collects data with Industrial Internet Reference Architecture model. It consists of edge level, the platform level, and the enterprise level. At the edge level, various industrial IIoT devices and protocols are deployed. The platform level encompasses cloud services and related infrastructure. At the enterprise level, dedicated machines or systems are utilized to launch and simulate cyberattacks targeting the IIoT environment. The dataset broadly classifies into nine attack types which is further classified into eighteen subattack types. Nine attack categories are: reconnaissance, weaponization, exploitation, lateral movement, command and control (C&C), crypto-ransomware, ransom DoS, exfiltration, and tampering. For more details about their attacks, please refer [37]. The dataset consists of 421 417 normal records, 399 417 attack records. Additionally, the dataset comprises 59 features extracted from various sources, including network traffic, device resources, and device/alert logs. Table V illustrates

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

TABLE V
TRAINING AND TEST SAMPLES OF X-IIoTID DATASET

Attack Category	Training Dataset	Test Dataset
Normal	252 850	84 283
Reconnaissance	76 554	25 518
Weaponization	40 356	13 452
Exploitation	679	226
Lateral movement	18 957	6319
C&C	1717	572
Exfiltration	13 280	4426
Tampering	3073	1024
Crypto-ransomware	274	91
Ransom DoS	84 756	28 252

distribution of normal and attack categories for training and test sets after data preprocessing.

The evaluation of the proposed intrusion detection model's performance is conducted by analyzing several key metrics: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The term TP indicate correctly predicted malicious data, TN indicates correctly predicted normal data, FP indicates normal data that is predicted as malicious data, and FN indicates malicious data misjudged as normal data. Evaluation metrics considered for assessment and validation of proposed scheme are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{31}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{32}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{33}$$

$$\text{F1} - \text{score} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})}. \tag{34}$$

2) ROC Curve: This curve offers a graphical representation of the model's ability to differentiate between different classes or categories. The area under the receiver operating characterstics (ROC) curve (AUC) quantifies the model's capability to distinguish between normal and attack instances. A higher AUC score indicates a more efficient and accurate model in separating the two categories.

a) Communication and computing models: The communication model employed in the experimental setup incorporates specific assumptions and configurations. The noise power N_0 is assumed to be –174 dBm, while the maximum transmitting power of each device is set to 30 dBm. The bandwidth $B_{\max}$ is designated as 100 MHz, and the channel gain $g_{i,u}$ is modeled using Rayleigh fading, with an average path loss of 10^{-3}. Additionally, the model parameters are quantized into 16 bits before transmission to optimize the communication process. Regarding the computing model, the CPU core frequency is set to 1100 MHz and $\varsigma_i = 20$ cycles/bit, ensuring adequate processing capabilities for the intrusion detection system and related computations.

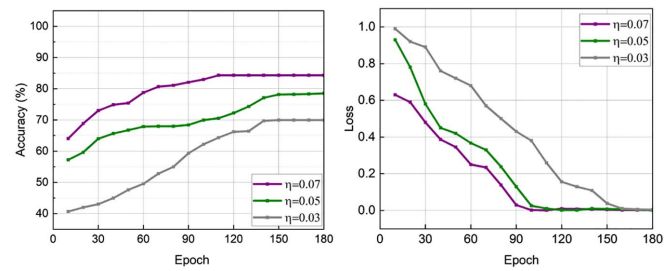

Fig. 3. Effect of hyperlearning rate (η).

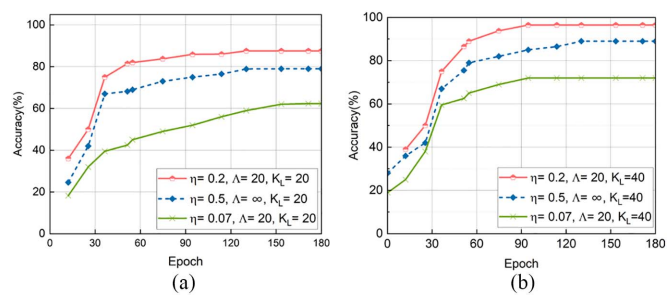

Fig. 4. Effect of different batch sizes, local computation rounds, and hyper-learning rate. (a) Effect of varying batch size and hyper-learning rate with KL = 20. (b) Effect of varying batch size and hyper-learning rate with KL = 40.

b) Baseline models: The performance of proposed intrusion detection models is compared with following FL and non-FL intrusion detection models.

i) *FL-DRL [30]*: Each participating device in the FL training process has assigned weights based on its attention value by varying the distribution of data from considered IIoT dataset.

ii) *FedAvg [10]*: All the device participating in the FL training process without accessing any criteria of data collected from these devices.

iii) *DRL [18]*: The features of each dataset are considered as the state and their labels as action to train the model. DRL training process is implemented to maximize the reward. This model is considered as non-FL model for evaluation.

B. Convergence Analysis

In this section, the various hyperparameters such as hyper-learning rate (η), discount factor (γ), batch size (Λ), and local computation round (K_L) are tuned and examined. We first examine the impact of changing hyperlearning rate (η) in Fig. 3(a) and 3(b) in terms of accuracy and loss with TON_IoT dataset (IID distribution). As can be seen in the figure, the proposed intrusion detection model converges even for small values of η. It is also observed that by increasing the value of η makes convergence better and achieves a maximum accuracy of 84.05% with $\eta = 0.07$. Further, the performance of the proposed model is examined under different settings of Λ,K_L, and η in Fig. 4(a) and 4(b) but consider similar setting of dataset. Fig. 4 demonstrates that with increasing the value of $\eta = 0.2$ improves convergence performance and also achieves an accuracy of 92.66%.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

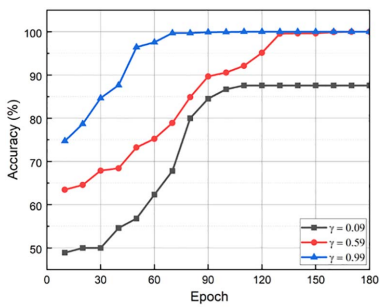

Fig. 5. Effect of discount factor.

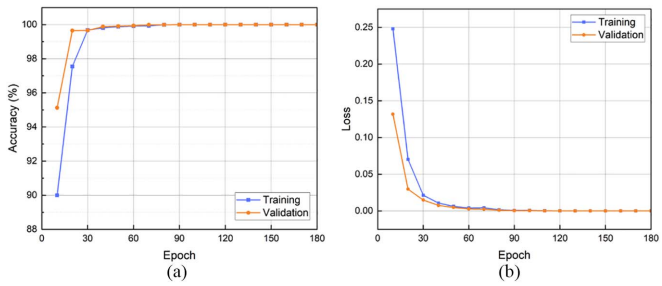

Fig. 6. (a) Accuracy. (b) Training loss of proposed intrusion detection model on TON_IoT dataset.

But further increase in the value of η, results in reduction in accuracy. It is also observed that using larger batch size benefits the stability but also makes the convergence slow. Specifically, when we increase K_L from 20 to 40, with proper tuning of hyperlearning rate and batch size settings, achieves a significant improvement in accuracy. However, large value of K_L requires more computation for local updates which increases the cost function in terms of energy consumption of each participating device. It is anticipated from Fig. 4 that the proposed model shows improvement in accuracy for different batch size Λ settings but with very large batch size ($\Lambda = \infty$) slows down the convergence of model. Fig. 5 shows the impact of discount factor on the performance of the proposed intrusion detection model. To evaluate the performance, the value of discount factors γ are set to be 0.09, 0.59, 0.99. It is observed that higher value of γ, obtains better accuracy. This is due to the reason that higher value of discount factor, concerned more about future rewards, thus guaranteed long-term rewards.

Fig. 6(a) and 6(b) shows the achieved accuracy and loss with TON_IoT dataset using proposed intrusion detection model. It is observed that the accuracy and loss of 99.95% and 0.0023% has been achieved, respectively. Furthermore, Fig. 7 reflects the area under ROC curve which demonstrates the effective detection capabilities of the proposed model.

C. Performance Comparison Using Different Datasets and FL Baseline schemes

To evaluate the performance of the proposed framework, experiments are conducted on public datasets which specifically considers the wide domain of intrusion attack categories in IoT/IIoT scenarios and compared with existing FL baseline intrusion detection models as illustrated in Table VI–VIII. As

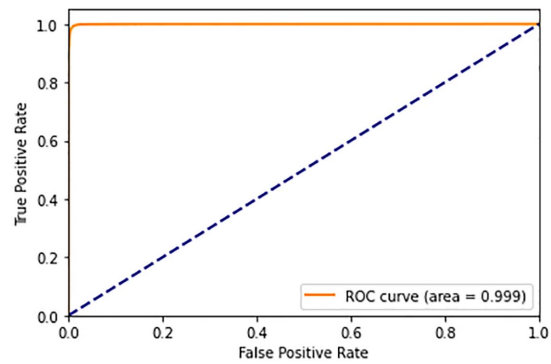

Fig. 7. ROC curve of proposed intrusion detection model on TON_IoT dataset.

described in Table VI, the proposed model outperforms as compared to baseline models using TON_IoT dataset. The accuracy achieved with the proposed model is 99.95%, precision is 98.63%, recall is 99.98%, and F1-score is 99.99%. In the literature, the researchers have presented various intrusion detection models with their prime focus on attack vectors and ignored their attention to enhance the performance of FL system while minimizing the energy consumption. The offered solution in this study successfully addresses these issues, not only detects the network intrusion but also enhances the performance of FL with the integration of DRL to select high quality client devices for global aggregation model.

Table VII illustrates the performance comparison of the proposed model with existing FL baseline models using Edge_IoT dataset. The accuracy achieved by our proposed model is 97.90% in multiclass classification. Also, high value of precision, recall and F1-score is also scored as compared to other baseline models. The performance of proposed and existing intrusion detection models is also examined using X-IIoTID dataset. It consists of nine attack subcategories which is accurately classified as multiclass classification with an accuracy of 99.99% whereas other baseline schemes achieve an accuracy of 90.24% and 81.35% under similar scenario.

In a nutshell, the proposed GRU-assisted Federated DRL intrusion detection model provides a robust and secure solution for IIoT systems with high threat detection accuracy.

D. Performance With IID and Non-IID Data

To validate the performance of the proposed model in real-world scenario, the data distribution needs to be non-IID and imbalanced. For experimental purpose, Egde-IIoT dataset (FL-specific dataset) have been divided into several local datasets to train them as per FL requirement. To examine the accuracy, the training dataset is shared among varying number of clients. Here, client size is considered as 3, 9, and 15, then the performance is evaluated for the first and fiftieth round of learning. Table IX compares the performance in terms of accuracy for global model, best clients, and worst clients for both IID and non-IID data distribution mode. It is observed that the performance for all classes increases with increase in number of rounds. In case of IID, there is small performance gap between best and worst clients. However, there is quite large gap for

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
PERFORMANCE COMPARISON USING TON_IoT DATASET

Model	Metric	Attack Categories										Macro Average (%)	Acc. (%)
		Normal	Scanning	DoS	DDoS	Ransomware	Backdoor	Injection	XSS	Password	MITM		
Proposed	Precision (%)	98.76	96.42	99.99	100.00	97.98	98.80	99.99	100.00	100.00	99.99	98.63	99.95
	Recall (%)	99.96	99.99	80.98	97.79	94.78	100.00	84.89	76.45	98.99	100.00	99.98	
	F1-score (%)	98.09	97.89	88.75	94.88	95.85	98.80	90.98	86.00	98.99	99.98	99.99	
FL-DRL	Precision (%)	97.78	73.88	66.42	94.63	90.67	97.98	67.78	100.00	97.45	94.67	87.95	84.25
	Recall (%)	96.42	92.78	90.77	97.88	76.65	48.57	87.93	94.67	84.00	97.98	86.00	
	F1-score (%)	96.07	81.42	76.55	95.66	82.04	64.00	75.76	96.98	90.87	95.04	85.45	
FedAvg	Precision (%)	96.87	44.06	0.00	97.90	42.75	0.00	0.00	100.00	100.00	76.66	55.63	68.07
	Recall (%)	87.09	99.87	0.00	59.00	47.88	0.00	0.00	86.02	17.56	99.99	49.00	
	F1-score (%)	91.33	60.42	0.00	73.67	44.67	0.00	0.00	92.00	29.42	85.78	47.00	

TABLE VII
PERFORMANCE COMPARISON USING EDGE_IIoT DATASET

Model	Metric	Attack Categories						Macro Average (%)	Acc. (%)
		Normal	Scanning	DDoS	Malware	Injection	MITM		
Proposed	Precision (%)	100.00	96.99	99.99	94.98	96.99	100.00	97.56	97.90
	Recall (%)	100.00	97.87	100.00	97.67	87.56	97.67	96.98	
	F1-score (%)	100.00	96.88	99.99	95.00	91.98	97.33	97.95	
FL-DRL	Precision (%)	100.00	92.05	98.56	93.42	63.00	99.56	90.09	89.50
	Recall (%)	99.87	82.31	83.02	73.44	86.32	100.09	87.56	
	F1-score (%)	99.87	86.77	89.00	81.34	72.86	99.56	87.34	
FedAvg	Precision (%)	94.00	76.66	73.52	97.56	61.65	96.34	82.56	78.33
	Recall (%)	98.09	80.33	93.00	48.54	57.00	98.56	79.34	
	F1-score (%)	95.87	77.69	81.56	64.70	58.67	97.34	78.55	

TABLE VIII
PERFORMANCE COMPARISON USING X-IIoTID DATASET

Model	Metric	Attack Categories										Macro Average (%)	Acc. (%)
		Normal	Reconnaissance	Weaponization	Exploitation	Lateral movement	C&C	Exfiltration	Tampering	Crypto-ransomware	Ransom DoS		
Proposed	Precision (%)	100.00	99.62	99.99	98.93	99.99	86.23	98.75	99.97	99.56	100.00	97.98	99.99
	Recall (%)	100.00	100.00	99.87	97.82	96.98	99.87	97.89	99.99	99.00	98.83	98.99	
	F1-score (%)	100.00	99.45	99.90	97.99	97.87	92.00	97.66	99.98	99.44	98.83	97.78	
FL-DRL	Precision (%)	99.65	87.58	99.56	98.95	98.22	63.55	87.92	99.09	99.76	94.44	92.67	90.24
	Recall (%)	99.34	95.21	100.00	99.00	34.55	97.92	52.01	100.00	45.00	86.00	80.87	
	F1-score (%)	99.66	89.99	99.56	98.77	5.78	76.54	65.00	99.43	61.34	89.87	82.50	
FedAvg	Precision (%)	99.33	87.86	95.92	93.65	78.62	51.07	83.21	88.45	82.33	98.00	85.00	81.35
	Recall (%)	97.00	91.92	83.00	97.88	67.52	58.52	99.94	63.56	80.11	90.14	82.20	
	F1-score (%)	98.83	88.56	88.52	94.43	72.77	54.00	90.31	73.00	80.56	93.65	82.65	

non-IID but the clients under FL setting are able to achieve better accuracy as the round increases. For an instance, for client size $i = 15$, the accuracy for worst client is 67.73% but with fiftieth round of learning, the worst client achieved an accuracy of 82.96%. This is due to the fact that during initial round of training, high variations in data quality across clients result in less effective local models, which aggregate less efficiently into a global model and results into lower initial accuracies. As the number of trainings round increases, the result indicates significant improvement in accuracy as GRUs handle the variability in temporal data patterns collected in previous rounds and DRL selection strategy maintains the overall quality and relevance of the data used in each training round.

E. Comparison With Other Baseline Models

Here, the performance of the proposed model is compared with FL as well as non-FL models. For this comparison, DRL

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
PERFORMANCE COMPARISON IN TERMS OF ACCURACY FOR IID AND NON-IID DISTRIBUTIONS

		First Round						Fiftieth Round					
		IID			Non-IID			IID			Non-IID		
Model	Clients	Best (%)	Worst (%)	Global (%)	Best (%)	Worst (%)	Global (%)	Best (%)	Worst (%)	Global (%)	Best (%)	Worst (%)	Global (%)
Proposed	i = 3	97.25	97.54	97.44	99.98	98.63	99.56	100	99.98	100	100	99.99	100
	i = 9	98.63	98.88	98.76	98.98	87.15	99.99	100	99.99	99.99	99.99	98.88	99.99
	i = 15	98.32	98.68	98.56	99.92	67.73	92.70	100	99.99	99.99	99.98	82.96	97.98
FL-DRL	i = 3	95.32	95.88	95.67	93.06	07.31	86.46	95.94	95.98	95.98	93.13	07.31	86.76
	i = 9	95.33	95.99	95.86	95.21	10.72	87.35	96.00	96.54	95.99	95.67	10.76	88.00
	i = 15	95.32	95.75	95.63	93.31	10.77	87.89	95.63	95.88	95.89	94.06	11.07	89.78
FedAvg	i = 3	93.33	92.90	93.02	91.44	03.06	77.04	93.62	92.99	93.59	91.77	39.89	89.45
	i = 9	92.97	93.32	93.11	91.13	08.23	73.78	93.86	93.87	93.45	92.52	11.28	82.05
	i = 15	92.32	93.02	93.46	88.19	10.77	71.42	93.38	93.91	93.37	90.83	12.45	89.62

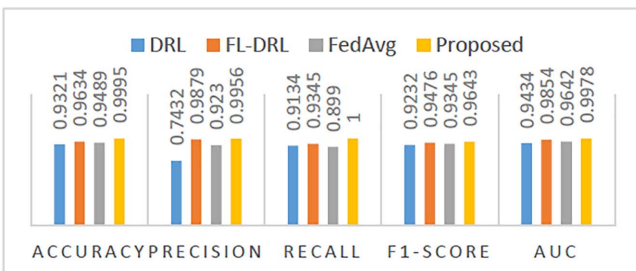

Fig. 8. Performance comparison of FL and non-FL models using TON_IoT.

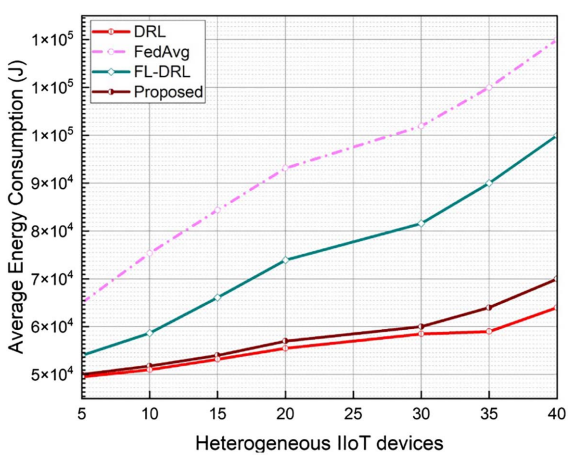

Fig. 9. Impact of increasing heterogeneous IIoT devices.

(as non-FL model) [18] is considered for intrusion detection. In [18], best features have been selected with random forest and decision tree classifiers from considered attack specific dataset and then samples are fed into DRL model for training and validation. Fig. 8 compares the performance of the proposed model with baseline intrusion detection models using TON_IoT dataset. As can be seen from the figure, it is noticed that proposed model outperforms even if compared with non-FL model. However, non-FL models achieve an accuracy of 87.25% with involved feature selection process but do not ensure secure

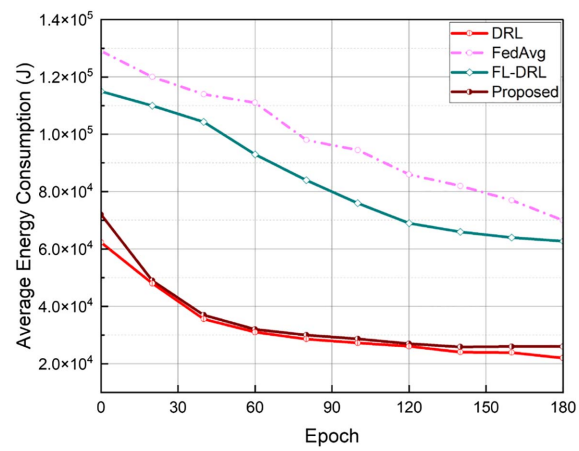

Fig. 10. Energy consumption with FL and non-FL models.

performance and high accuracy rates in classifying malicious attacks. Fig. 9 compares the performance in terms of average energy consumption with increasing number of heterogeneous IIoT devices. It is observed that the average energy consumption of the proposed model is less as compared to FL baseline models. This is due to the reason that the proposed model selects high-quality local nodes for participation in FL training rounds whereas other baselines schemes do not consider this fact and hence result in higher energy consumption. It is also demonstrated that the average energy consumption by the proposed model is slightly higher as compared to traditional DRL model for intrusion detection. The reason behind is that the DRL does not involves added communication, synchronization, and privacy overhead associated with FL approaches.

As illustrated in Fig. 10, the proposed model consistently shows lower energy consumption compared to FL-DRL and FedAvg with respect to training epochs. This is the direct result of improved learning with GRU layers and high-quality device selection for efficient learning and fewer training rounds. However, energy consumption by our proposed model is slightly higher than DRL model as it incurs additional communication and synchronization process.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

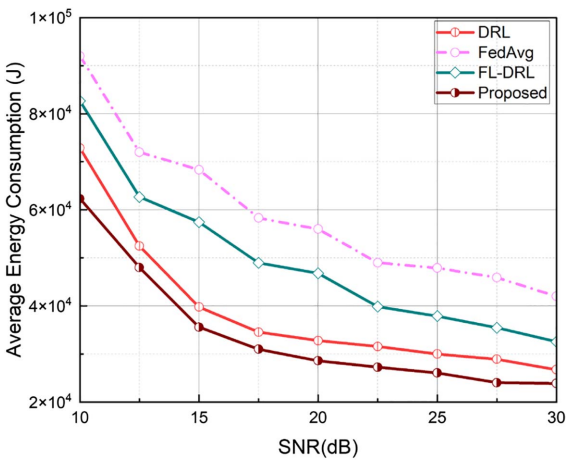

Fig. 11. Impact of channel conditions on energy consumption.

Fig. 11 illustrates the impact of varying SNR during to dynamic channel conditions and its impact on energy consumption. As seen from Fig. 11, the proposed model demonstrates the most efficient energy consumption reduction as compared to baseline models. The is due to the reason that proposed model incorporates GRUs which enhance the learning efficiency by capturing temporal dependencies efficiently and thus reduce energy consumption. Also, by integrating DRL, only the nodes having good SNR values are selected for FL training process. This results in reduced energy consumption of proposed framework. However, other FL models might be better optimized for the specific constraints and requirements of IIoT environment.

To sum up, through the analysis of comparison with FL and non-FL, it can be concluded that the proposed model provides secure comprehensive framework for intrusion detection along with reduced energy consumption in industrial environment.

VI. Conclusion and Future Directions

In this article, we introduced a GRU-assisted federated deep recurrent reinforcement learning-based intrusion detection model to develop robust and secure framework against poisoning adversarial attacks for IIoT scenarios. The implementation of GRU-assisted FL part of the proposed framework concentrates on security aspect. It provides secure model with data privacy by keeping data intact to local models and also detecting the intrusion in incoming network traffic by mapping their temporal dependencies. The integration of DRL with above framework improves the model training efficiency and accuracy by selecting high quality IIoT devices for data aggregation with reduced communication cost. The performance of the proposed intrusion detection model has been evaluated and validated with industry specific datasets which considers recent and specific malicious attacks in IIoT domain and also consider non-IID data distribution to verify its effectiveness in real-world scenarios. The experimental results validate that the proposed model outperforms the non-FL and other variants of FL for intrusion detection. As a future work, we will analyze and evaluate the performance of the proposed model with real-time data from device-specific datasets, to detect security vulnerabilities of IIoT device due to malicious attacks.

References

[1] R. Lohiya and A. Thakkar, "Application domains, evaluation data sets, and research challenges of IoT: A systematic review," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 8774–8798, Jun. 2021.

[2] S. Kumar, P. Tiwari, and M. Zymbler, "Internet of things is a revolutionary approach for future technology enhancement: A review," *J. Big Data*, vol. 6, no. 1, pp. 1–21, 2019.

[3] D. Borsatti, G. Davoli, W. Cerroni, and C. Raffaelli, "Enabling industrial IoT as a service with multi-access edge computing," *IEEE Commun. Mag.*, vol. 59, no. 8, pp. 21–27, Aug. 2021.

[4] E. Sisinni, A. Saifullah, S. Han, U. Jennehag, and M. Gidlund, "Industrial Internet of Things: Challenges, opportunities, and directions," *IEEE Trans. Ind. Informat.*, vol. 14, no. 11, pp. 4724–4734, Nov. 2018.

[5] "Industrial IoT: Future market outlook, technology analysis and key players 2020-2025." Juniper Research. Accessed: Sep. 24, 2023. [Online]. Available: https://www.juniperresearch.com/press/industrial-iot-iiot-connections-smart-factories

[6] R. Kumar, B. Kandpal, and V. Ahmad, "Industrial IoT (IIoT): Security threats and countermeasures," in *Proc. Int. Conf. Innov. Data Commun. Technol. Appl. (ICIDCA)*, Uttarakhand, India, 2023, pp. 829–833.

[7] A. R. Sadeghi, C. Wachsmann, and M. Waidner, "Security and privacy challenges in industrial Internet of Things," in *Proc. 52nd ACM/EDAC/IEEE Des. Automat. Conf. (DAC)*, San Francisco, CA, USA, 2015, pp. 1–6.

[8] Z. Ji, Z. C. Lipton, and C. Elkan, "Differential privacy and machine learning: A survey and review," 2014. [Online]. Available: https://arxiv.org/abs/1412.7584

[9] D. C. Nguyen et al., "Federated learning for industrial Internet of Things in future industries," *IEEE Wireless Commun.*, vol. 28, no. 6, pp. 192–199, Dec. 2021.

[10] P. Ruzafa-Alcázar et al., "Intrusion detection based on privacy-preserving federated learning for the industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1145–1154, Feb. 2023.

[11] W. Yang, W. Xiang, Y. Yang, and P. Cheng, "Optimizing federated learning with deep reinforcement learning for digital twin empowered industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1884–1893, Feb. 2023.

[12] Q. Wu, K. He, and X. Chen, "Personalized federated learning for intelligent IoT applications: A cloud-edge based framework," *IEEE Open J. Comput. Soc.*, vol. 1, pp. 35–44, 2020.

[13] P. Zhang, C. Wang, C. Jiang, and Z. Han, "Deep reinforcement learning assisted federated learning algorithm for data management of IIoT," *IEEE Trans. Ind. Informat.*, vol. 17, no. 12, pp. 8475–8484, Dec. 2021.

[14] N. H. Nguyen et al., "FedDRL: Deep reinforcement learning-based adaptive aggregation for non-IID data in federated learning," in *Proc. 51st Int. Conf. Parallel Process.*, 2022, pp. 1–11.

[15] T. G. Nguyen, T. V. Phan, D. T. Hoang, T. N. Nguyen, and C. So-In, "Federated deep reinforcement learning for traffic monitoring in SDN-based IoT networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 4, pp. 1048–1065, Dec. 2021.

[16] L. Vu, V. L. Cao, Q. U. Nguyen, D. N. Nguyen, D. T. Hoang, and E. Dutkiewicz, "Learning latent representation for IoT anomaly detection," *IEEE Trans. Cybern.*, vol. 52, no. 5, pp. 3769–3782, May 2022.

[17] J. Long, W. Liang, K.-C. Li, Y. Wei, and M. D. Marino, "A regularized cross-layer ladder network for intrusion detection in industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1747–1755, 2023.

[18] M. Hasan, M. M. Islam, M. I. I. Zarif, and M. M. A. Hashem, "Attack and anomaly detection in IoT sensors in IoT sites using machine learning approaches," *Internet Things*, vol. 7, Sep. 2019, Art. no. 100059.

[19] H. Lin, J. Hu, X. Wang, M. F. Alhamid, and M. J. Piran, "Toward secure data fusion in industrial IoT using transfer learning," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 7114–7122, Oct. 2021.

[20] K. Ren, Y. Zeng, Z. Cao, and Y. Zhang, "ID-RDRL: A deep reinforcement learning-based feature selection intrusion detection model," *Sci. Rep.*, vol. 12, no. 1, 2022, Art. no. 15370.

[21] Y. Zhang, C. Yang, K. Huang, and Y. Li, "Intrusion detection of industrial Internet-of-Things based on reconstructed graph neural networks," *IEEE Trans. Netw. Sci.*, vol. 10, no. 5, pp. 2894–2905, Sep./Oct. 2023.

[22] Y. Li et al., "Robust detection for network intrusion IoT based on multi-CNN fusion," *Measurement*, vol. 154, 2020, Art. no. 107450.

[23] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.

[24] A. Halbouni, T. S. Gunawan, M. H. Habaebi, M. Halbouni, M. Kartiwi, and R. Ahmad, "CNN-LSTM: Hybrid deep neural network for network intrusion detection system," *IEEE Access*, vol. 10, pp. 99837–99849, 2022.

[25] X. Li, M. Xu, P. Vijayakumar, N. Kumar, and X. Liu, "Detection of low-frequency and multi-stage attacks in industrial Internet of Things," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 8820–8831, Aug. 2020.

[26] F. Kong, J. Li, B. Jiang, H. Wang, and H. Song, "Integrated generative model for industrial anomaly detection via bidirectional LSTM and attention mechanism," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 541–550, Jan. 2023.

[27] S. Latif, Z. Idrees, Z. Zou, and J. Ahmad, "DRaNN: A deep random neural network model for intrusion detection in industrial IoT," in *Proc. Int. Conf. UK-China Emerg. Technol. (UCET)*, Glasgow, U.K., 2020, pp. 1–4.

[28] J. Liu et al., "Toward security monitoring of industrial cyber-physical systems via hierarchically distributed intrusion detection," *Expert Syst. Appl.*, vol. 158, Aug. 2020, Art. no. 113578.

[29] X. Wang et al., "Toward accurate anomaly detection in industrial internet of things using hierarchical federated learning," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7110–7119, Oct. 2022.

[30] B. Li, Y. Wu, J. Song, R. Lu, T. Li, and L. Zhao, "DeepFed: Federated deep learning for intrusion detection in industrial cyber–physical systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5615–5624, Aug. 2021.

[31] Z. Chen, N. Lv, P. Liu, Y. Fang, K. Chen, and W. Pan, "Intrusion detection for wireless edge networks based on federated learning," *IEEE Access*, vol. 8, pp. 217463–217472, 2020.

[32] B. Weinger et al., "Enhancing IoT anomaly detection performance for federated learning," *Digit. Commun. Netw.*, vol. 8, no. 3, pp. 314–323, 2022.

[33] Y. Liu et al., "Deep anomaly detection for time-series data in industrial IoT: A communication-efficient on-device federated learning approach," *IEEE Internet Things J.*, vol. 8, no. 8, pp. 6348–6358, Aug. 2021.

[34] S. Vadigi, K. Sethi, D. Mohanty, S. P. Das, and P. Bera, "Federated reinforcement learning based intrusion detection system using dynamic attention mechanism," *J. Inf. Secur. Appl.*, vol. 78, Oct. 2023, Art. no. 103608.

[35] C. T. Dinh et al., "Federated learning over wireless networks: Convergence analysis and resource allocation," *IEEE/ACM Trans. Netw.*, vol. 29, no. 1, pp. 398–409, Feb. 2021.

[36] R. Chen, L. Li, K. Xue, C. Zhang, M. Pan, and Y. Fang, "Energy efficient federated learning over heterogeneous mobile devices via joint design of weight quantization and wireless transmission," *IEEE Trans. Mobile Comput.*, vol. 22, no. 12, pp. 7451–7465, Dec. 2023.

[37] A. Kaur and K. Kumar, "A reinforcement learning-based green resource allocation for heterogeneous services in cooperative cognitive radio networks," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 2, pp. 1554–1566, Jun. 2022.

[38] A. Kaur, K. Kumar, A. Praksah, and R. Tripathi, "Imperfect CSI based resource management in cognitive IoT networks: A deep recurrent reinforcement learning framework," *IEEE Trans. Cogn. Commun. Netw*, vol. 9, no. 5, pp. 1271–1281, Oct. 2023.

[39] Y. Liu, J. J. Q. Yu, J. Kang, D. Niyato, and S. Zhang, "Privacy-preserving traffic flow prediction: A federated learning approach," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7751–7763, Aug. 2020.

[40] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, "On the convergence of FedAvg on non-IID data," 2019, *arXiv:1907.02189*.

[41] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40281–40306, 2022.

[42] M. Al-Hawawreh, E. Sitnikova, and N. Aboutorab, "X-IIoTID: A connectivity-agnostic and device-agnostic intrusion data set for industrial Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 5, pp. 3962–3977, Mar. 2022.

[43] X. Wang et al., "AI-empowered trajectory anomaly detection for intelligent transportation systems: A hierarchical federated learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4631–4640, Apr. 2023.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on March 03,2026 at 04:11:52 UTC from IEEE Xplore. Restrictions apply.